

Oral virome in periodontal health and periodontitis

Reanalysis of PRJNA1183294 subgingival shotgun metagenomes. Portfolio pipeline, not a clinical finding.

Question

In subgingival gingival crevicular fluid, does the viral fraction of the metagenome differ in diversity or composition between periodontally healthy subjects and periodontitis patients?

Methods

16 samples were taken from BioProject PRJNA1183294 (Periodontally healthy n=8, Periodontitis n=8), balanced across 4 countries (Belgium, Chile, Peru, Spain), and subsampled to 1,000,000 read pairs with seqtk at seed 42 so that depth is identical across samples and the pipeline runs on a laptop. Reads were adapter and quality trimmed with fastp, with FastQC before and after and a MultiQC summary. Surviving pairs were aligned with bowtie2 to NCBI RefSeq viral genomes (release 237), keeping only properly paired alignments at MAPQ ≥ 30 ; a virus counted as present only with ≥ 10 reads across $\geq 1\%$ of its genome, a breadth rule that suppresses pileups on single repetitive loci. Alpha diversity was compared with exact Wilcoxon rank-sum tests and Bray-Curtis composition with PERMANOVA (9999 permutations) in R with vegan. Human host reads were not depleted, which is a stated limitation.

Table 1. Read retention and viral mapping

Sample	Group	Subsampled pairs	Post-QC pairs	QC %	Viral pairs	Viral %
BEHE003	Healthy	1,000,000	955,931	95.6	69	0.0072
BEHE007	Healthy	1,000,000	873,708	87.4	74	0.0085
CLHE003	Healthy	1,000,000	961,365	96.1	45	0.0047
CLHE005	Healthy	1,000,000	955,313	95.5	57	0.0060
ESHE017	Healthy	1,000,000	914,841	91.5	73	0.0080
ESHE020	Healthy	1,000,000	964,098	96.4	54	0.0056
PEHE004	Healthy	1,000,000	938,352	93.8	58	0.0062
PEHE005	Healthy	1,000,000	954,103	95.4	52	0.0055
BEPE001	Periodontitis	1,000,000	970,199	97.0	44	0.0045
BEPE002	Periodontitis	1,000,000	932,811	93.3	48	0.0051
CLPE012	Periodontitis	1,000,000	937,571	93.8	56	0.0060
CLPE014	Periodontitis	1,000,000	914,390	91.4	48	0.0052
ESPE023	Periodontitis	1,000,000	944,395	94.4	46	0.0049
ESPE024	Periodontitis	1,000,000	888,768	88.9	35	0.0039
PEPE010	Periodontitis	1,000,000	943,358	94.3	29	0.0031
PEPE022	Periodontitis	1,000,000	938,457	93.8	20	0.0021
TOTAL		16,000,000	14,987,660	93.7	808	0.0054

Results

4 RefSeq viral sequences passed the presence filter in at least one sample. Viral read pairs were 0.0054% of post-QC pairs.

Observed richness: median 2.00 (Periodontally healthy) vs 1.00 (Periodontitis); $W = 20.0$, $p = 0.1607$, rank-biserial $r = -0.38$.

Shannon: median 0.41 (Periodontally healthy) vs 0.00 (Periodontitis); $W = 23.0$, $p = 0.3253$, rank-biserial $r = -0.28$.

PERMANOVA bray ~ diagnosis, term diagnosis: $R^2 = 0.017$, $F = 0.24$, $p = 0.9816$.

PERMANOVA bray ~ country + diagnosis, term country: $R^2 = 0.266$, $F = 1.36$, $p = 0.2712$.

PERMANOVA bray ~ country + diagnosis, term diagnosis: $R^2 = 0.017$, $F = 0.25$, $p = 0.9692$.

Dispersion homogeneity (betadisper): $F = 0.80$, $p = 0.3652$.

Most aligned reads fall on a small number of references: Human endogenous retrovirus K113 (1148 reads in 16/16 samples); UNVERIFIED_ORG: Proteus phage VB_PmIS-Isfahan (304 reads in 16/16 samples); Human papillomavirus type 4 (40 reads in 1/16 samples); BeAn 58058 virus (32 reads in 10/16 samples). The top two account for 90% of all aligned reads. The first is a human endogenous retrovirus, which is host genomic sequence rather than an exogenous virus, and the second is a reference NCBI itself flags UNVERIFIED_ORG. Read the headline counts as an upper bound dominated by artefact, not as a virome.

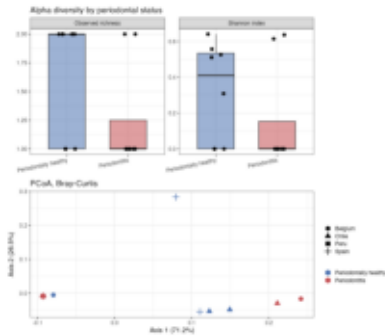


Figure 1. Alpha diversity by group, and Bray-Curtis PCoA.

Limitations

- Cross-sectional data. Any difference is an association, not evidence that either condition causes the other.
- Each sample was subsampled to 1,000,000 read pairs to run on a laptop, discarding most of the data and lowering sensitivity to rare viruses.
- This is a bulk shotgun metagenome, not a VLP-enriched virome, so viral reads are a small and mostly bacteriophage fraction and counts are sparse.
- Human reads were not depleted. Host sequence can align spuriously to viral references, so absolute viral abundance should be read as an upper bound.
- $n = 16$ across 4 countries. The source study found country explained more compositional variance than diagnosis, and this subset cannot separate the two.
- Detection is reference-based, so viruses absent from RefSeq viral, which is much of the oral phage population, are invisible here.